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Curators or creators: Role configurations of digital innovation strategy in museum tourism destination and the principles underlying their attractiveness

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ABSTRACT

Digital innovation has emerged as a pivotal strategy for enhancing museum tourism attractiveness, yet its enablers and complex mechanisms are still underexplored. Should museums enhance existing experiences through exploitative innovation or create novel ones through exploratory innovation? This research selects 31 provincial representative museums in China as samples and undertakes a comprehensive case analysis employing interviews, official documents, public records, and other relevant materials. The analysis combines case analysis with fuzzy-set Qualitative Comparative Analysis (fsQCA). Our findings reveal: (1) Critical enablers include cultural relics data assets, smart tourism systems, dynamic capabilities, innovation alliances, public demand, and financial backing. (2) Through various configurations of enablers, we reveal three approaches: Extroverted Creator (exogenous-oriented exploratory), Responsive Curator (exploitative), and Introverted Creator (endogenous-oriented exploratory). Results deepen theoretical insights into digital innovation in museum tourism and equip destinations with a strategic positioning map to assess their own innovation ecosystem and develop context-specific strategies.

1. Introduction

In a mere square inch, a millennium unfolds. Museums, as prominent cultural tourism destinations globally, are currently experiencing rapid advancements in both tourism and digital development (Xu et al., 2024; He et al., 2018; Tham, Liu, & Loo, 2023; Tim, Ouyang, & Zeng, 2020; Trunfio, Lucia, Campana, & Magnelli, 2022). Digital innovation has become essential for enhancing tourist experiences and boosting tourism attractiveness and competitiveness (Giannini & Bowen, 2022; Palumbo, 2022; Taormina & Baraldi, 2022; Zollo, Rialti, Marrucci, & Ciappei, 2022). On the one hand, museums enhance existing exhibits and visitor interactions through exploitative digital innovation (Ch'ng, Cai, Leow, & Zhang, 2019). On the other hand, they create new offerings through exploratory digital innovation to meet niche visitor demands (Taormina & Baraldi, 2022). Both types of innovation are significant, yet each has its limitations. Exploitative innovation often yields short-term gains but is less likely to induce profound changes in tourism patterns (O'Reilly & Tushman, 2004). Conversely, exploratory innovation can transcend traditional boundaries of museum attractiveness and competitiveness but entails greater uncertainty and delayed returns (O'Reilly & Tushman, 2004). For museums that specialize in cultural preservation rather than tourism and technology, deciding between exploitative innovation and exploratory innovation remains a challenge (Wang, Li, Ruan, Zhang, & Li, 2024). To better understand these strategies, it is essential to clarify the multidimensional enablers associated with them.

Current research on digital innovation in museum tourism primarily centers on technologies enhancing visitor experiences, including mobile applications, Virtual Reality (VR), the Metaverse, and social media (Buhalis, Leung, & Lin, 2023; He et al., 2018). Additionally, managerial aspects such as big data, artificial intelligence, and machine learning are also extensively studied (Giannini & Bowen, 2022; Wang, Duan, Yang, Cao, & Pan, 2023). Organizational factors, including human resource capabilities and the willingness to adopt new technologies, are considered significant (Leoni & Cristofaro, 2022). However, the interplay between these elements and both exploitative and exploratory innovation strategies remains ambiguous. Furthermore, most existing studies focus on specific innovation scenarios (Tham et al., 2023) without adopting a comprehensive developmental perspective to identify the most critical

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0261-5177/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

enablers for advancing digital innovation in museums. This critical gap in the literature needs to be addressed.

Following the identification of key enablers, it is crucial to explore how these factors illuminate museum innovation strategies. According to the complex innovation systems perspective (Poutanen, Soliman, & Ståhle, 2016), innovation results from the nonlinear interaction of diverse elements, including technology, capabilities, knowledge, and assets (Dougherty & Hardy, 1996; Garud, Gehman, & Kumaraswamy, 2011). Organizations strategically orchestrate these resources to navigate innovation pathways (Dougherty, 2017). Therefore, it is essential to study how these enablers synergistically impact museums' distinct exploitative or exploratory digital innovation strategies, rather than analyzing their effects in isolation.

Drawing from the preceding analysis, this study is organized around two sequential research questions:

Study I: What are the key enablers influencing exploitative and exploratory digital innovation strategies in museum tourism? To address this question, we selected 31 leading provincial museums in China, recognized for their innovative leadership. Data were collected through interviews, official documents, and publicly available materials from these institutions. A comprehensive case analysis was conducted to explore the research question. The Technology-Organization-Environment (TOE) framework significantly informed our analysis. This framework, widely utilized in studies of organizational innovation behaviors, offers a comprehensive and balanced perspective by integrating technological, organizational, and environmental enablers (Tornatzky & Fleischer, 1990).

Study II: How do these enablers collectively shape distinct exploitative or exploratory digital innovation strategies, in turn influencing the attractiveness of museum tourism? We address this question employing fuzzy-set Qualitative Comparative Analysis (fsQCA) on the same sample. FsQCA, which embodies configurational thinking, posits that enablers impact outcomes through diverse interactions and combinations (Ragin, 2009). This approach is capable of revealing the causal complexity between digital innovation enablers and various innovation strategies. Additionally, fsQCA assesses the consistency of causal configurations with outcomes (Fiss, 2011), identifying key digital innovation strategies that enhance the attractiveness of museum tourism.

2. Theoretical background

2.1. Digital innovation in museum tourism destinations: exploitative and exploratory strategies

Digital innovation, defined as leveraging digital technologies to drive innovation (Yoo, Henfridsson, & Lyytinen, 2010), is a crucial means of enhancing the attractiveness of museum tourism (García-Muiña, Fuentes-Moraleda, Vacas-Guerrero, & Rienda-Gómez, 2019; He et al., 2018; Pesce, Neirotti, & Paolucci, 2019; Tham et al., 2023). It not only facilitates the innovative optimization of existing artifact displays and visitor experiences but also has the potential to create disruptive museum tourism experiences. In this context, museums face the strategic dilemma of choosing between steady, incremental improvements and bold, pioneering transformations. To understand these two innovation strategies, this study references March (1991) and Benner and Tushman (2003), categorizing them as exploitative innovation and exploratory innovation. The former focuses on improving existing products, services, or processes, while the latter involves developing new products, services, or business models in uncharted territories.

Specifically, exploitative digital innovation in museum tourism seeks to optimize existing offerings through digital technologies, thereby maximizing the attractiveness and value proposition of attractions. Despite the increasing diversity in museum experience formats, many institutions maintain a strong focus on enhancing and innovating the presentation of physical artifacts, underscoring their dedication to this essential mission (Tham et al., 2023). As more visitors come to view cultural sites as integral to their recreational activities (Fernandez-Lores, Crespo-Tejero, & Fernández-Hernández, 2022), museums have augmented physical exhibits with informational displays and immersive technologies to enhance visitor engagement and interest (Raimo, De Turi, Ricciardelli, & Vitolla, 2022). For instance, interactive displays and Augmented Reality (AR) devices are utilized to enhance visitor engagement and memorability (Bec et al., 2019; He et al., 2018; Lee, Jung, tom Dieck, & Chung, 2020; Su and Teng, 2018). Additionally, big data analytics and electronic guide maps are employed to optimize the flow of exhibitions (Torres-Ruiz et al., 2020).

These innovations often receive prompt feedback regarding their impact on tourism attractiveness (O'Reilly & Tushman, 2004). However, focusing solely on traditional exhibition enhancements through exploitative innovation may not result in significant transformations in tourism models. Integrating experiential elements continues to pose challenges. Nevertheless, for museums committed to the traditional mission of artifact display or those cautiously embracing digitalization, such a strategy can effectively mitigate the risks associated with innovation (Tang, 2014).

In contrast, exploratory digital innovation in museum tourism promotes experimental approaches to developing new products, business models, and service delivery systems (García-Muiña et al., 2019; Giannini & Bowen, 2022). Given the increasingly diversified and personalized nature of visitor consumption, exploratory strategies can address niche demands and secure distinct advantages (Taormina & Baraldi, 2022). For instance, to cater to the leisure preferences of younger visitors, museums can develop gamified experiences, such as the Tate Time Machine at the Tate Museum (Alshawaaf & Lee, 2021; Wei, Zhang, Huang, & Qiu, 2023). For cultural enthusiasts, museums can develop blockchain-based digital collectibles and co-create content on digital platforms dedicated to cultural heritage (Pesce et al., 2019; Tham et al., 2023).

Successful exploratory innovation can significantly enhance visitor experiences and generates new revenue streams (Alshawaaf & Lee, 2021). For small and medium-sized museums with limited resources, strategically addressing emerging visitor demands can establish a distinctive digital advantage (Leoni & Cristofaro, 2022). However, these exploratory activities involve uncertainties and deferred returns (O'Reilly & Tushman, 2004), necessitating that museums demonstrate enhanced resource flexibility and elevated risk tolerance.

To achieve comprehensive cultural tourism development goals, museums may simultaneously adopt both exploitative and exploratory innovation strategies (Wang et al., 2024). In this process, exploratory innovation helps overcome the technological barriers posed by exploitative innovation, while the technology and knowledge accumulated through exploitative innovation can enhance exploratory efforts (O'Reilly & Tushman, 2004; Wang et al., 2024). However, the diverse demands for resources, talent, and funding for these strategies make it challenging for museums to maintain a balance (Leoni & Cristofaro, 2022). Consequently, museums may prioritize one innovation strategy over the other at different times.

2.2. Enablers of museum tourism digital innovation

Among the enablers of museum digital innovation, particular emphasis is placed on technological enablers and the transformations they induce (Lu, Moyle, Reid, Yang, & Liu, 2023). Technologies that enhance visitor experiences, such as mobile applications (Tham et al., 2023), social media (Agostino, Arnaboldi, & Lampis, 2020; Fu, Liu, & Li, 2024), AR, VR (He et al., 2018; Lin, Huang, & Ho, 2020), and the metaverse (Buhalis et al., 2023), play a crucial role. Additionally, technologies related to museum management, such as big data, artificial intelligence, and machine learning, enable the development of more sophisticated digital products through intelligent management and

decision-making processes (Giannini & Bowen, 2022; Wang et al., 2023).

Research in organizational studies suggests that the successful implementation of technology is contingent upon its alignment with the organization's specific characteristics (Tim et al., 2020). Given that most museum staff may lack expertise in both tourism management and digital innovation, human resources become a critical factor (Leoni & Cristofaro, 2022). Therefore, it is essential to cultivate an innovation ecosystem (Tham et al., 2023) that comprises internal digital professionals and multidisciplinary management teams (Taormina & Baraldi, 2022; Trunfio et al., 2022), along with support from external innovation actors with expertise in technology and cross-domain knowledge (Li & Ghirardi, 2019). Furthermore, the organization's readiness to adopt technology (Palumbo, 2022) and its size (Li & Ghirardi, 2019) are also critical considerations.

In addition to technological and organizational factors, several other influential elements have been sporadically discussed in the literature. Nearly all studies highlight the critical role of funding, whether sourced from public funds (Camarero, Garrido, & Vicente, 2011) or commercial enterprises (Johnstone, Bryce, & Alexander, 2023). The policy environment is another significant factor impacting museum innovation performance (Vicente, Camarero, & Garrido, 2012), especially since museums are often regulated by governmental agencies (Xu, Zhang, Zhang, Mao, & Wang, 2024). Furthermore, Leoni and Cristofaro (2022) emphasize the influence of market, economic, and cultural contexts. However, these factors have not yet received adequate attention in existing research.

Although the enablers of museum digital innovation have been extensively summarized, several research gaps remain. Firstly, most existing studies predominantly focus on specific innovation scenarios (Tham et al., 2023). In contrast, García-Muiña et al. (2019) argue that innovation should not be seen as a singular event but rather from a macro developmental perspective, emphasizing key enablers that support the sustainable growth of digital innovation in museums. Secondly, the interaction between these enablers and the strategies of exploitative and exploratory innovation warrants further investigation.

2.3. Complex innovation systems perspective on museum tourism digital innovation

The perspective of complex innovation systems offers valuable insights into the intricacies of digital innovation strategies within museum tourism, helping us understand their inherent complexity. This perspective posits that innovation arises from the nonlinear interactions among various elements—technology, capabilities, knowledge, and assets—rather than merely being the result of their individual contributions or a straightforward aggregation (Dougherty & Hardy, 1996; Garud et al., 2011; Poutanen et al., 2016). Consequently, organizations must strategically allocate their resources to ensure the feasibility of innovation (Dougherty, 2017). In cultural heritage destinations, factors such as environmental dynamism, innovation management, multiple stakeholders, innovation capabilities, and capital form an interconnected whole that collectively influences innovation outcomes (Wang et al., 2024). Nevertheless, current research has yet to elucidate how the complex interplay of digital innovation enablers shapes exploitative and exploratory innovation strategies in museums.

In addition, from the perspective of complex innovation systems, varied resource allocation methods can lead to a wide range of innovation trajectories (Dougherty, 2017; Garud et al., 2011). This suggests that various combinations of factors facilitating digital innovation in museums can lead to a spectrum of exploitative and exploratory strategies. Although previous case studies with small sample sizes have documented the innovation behaviors of specific museums (Alshawaaf & Lee, 2021), they have not systematically explored the variety of innovation models. Addressing this gap is essential for understanding the broader implications of innovation strategies. Furthermore, given the impracticality of exploring all possible innovation paths, it is crucial to employ appropriate research methodologies to identify the innovation strategies that most effectively enhance the attractiveness of museum tourism.

3. Research design

3.1. Research framework

This study develops a research framework that progresses from identifying enablers to recognizing enabler configurations, encompassing two sequential study phases:

Study phase I: Identification of enablers through case analysis.

Study phase II: Identification of enabler configurations through fsQCA analysis.

In Study I, a case analysis methodology is employed to identify the key enablers of exploitative and exploratory digital innovation. Subsequently, Study II utilizes fsQCA to examine these enablers, identified in Study I as antecedent conditions, to reveal from a configurational perspective how they collectively contribute to distinct models of exploitative or exploratory innovation. Given the sequential nature of these research phases, both studies must be based on the same sample, which should be suitable for both methodologies. Additionally, the number of conditions and the sample size derived from the case analysis in Study I must be balanced to meet the requirements of fsQCA (Ragin, 2009).

3.2. Research sample

According to the Chinese Ministry of Culture and Tourism, by 2022, China had 6565 museums, ranking among the countries with the highest number of museums globally. Over the past decade, the Chinese government has vigorously promoted the digitalization of museum tourism (-Ch'ng et al., 2019; Wang et al., 2023; Xu, Zhang, et al., 2024). Innovations such as smart guides, immersive experiences, digital performances, and gamified interactions have enriched physical visits. Concurrently, online services like virtual exhibitions, live streaming, and digital collections have facilitated remote interaction and learning (Liu, 2022). These innovative efforts have significantly enhanced the attractiveness of cultural tourism in Chinese museums, resulting in a museum boom. Consequently, China provides an ideal context for this research.

To determine our sample, we implemented a two-step process. Initially, our research encompassed all 31 provincial-level administrative regions in mainland China, ensuring the comprehensive applicability of our findings. Secondly, to gather insightful data, we focused on the uniquely representative provincial museum in each administrative region, which are typically named using the format "Region + Museum". These museums are distinguished by their significant cultural influence and leadership in innovation. Consequently, our research sample includes 31 representative provincial museums, labeled as M_1 through M_{31} (refer to Appendix 1).

For case analysis, 1) the selected museums—leaders in digital innovation within the tourism sector—offer invaluable practical insights; 2) the sample encompasses various regions across mainland China, where differences in digital innovation capabilities due to regional and developmental disparities provide richer comparative perspectives and more universally applicable research findings (Ch'ng et al., 2019). For the fsQCA, 1) this sample meets the diversity and comparability requirements; 2) as a medium-sized sample (10–40 cases), it allows for in-depth analysis and sufficient explanatory power to develop practical theories (Ragin, 2009).

3.3. Data collection

To maintain the highest level of rigor in our research, we employed a

multi-channel data collection approach and utilized triangulation methods (Yin, 1994). The data collection procedure is detailed in Table 1.

Secondary data collection. This approach enables a swift grasp of phenomena and establishes a foundational knowledge of digital innovation in museums. Since May 2021, our research team has conducted an extensive search for relevant information across official museum media, literature databases, and major media outlets. The search utilized keywords such as "museum name" combined with terms like "information", "digital", "data", and "smart" to gather pertinent data.

Primary data collection. To elucidate two critical aspects—specifically, the exploitative and exploratory digital innovation strategies utilized in representative digital tourism offerings by museums over the past three years, and the key enablers associated with these strategies—primary data was systematically collected. Since 2021, supported by the National Social Science Foundation, we have been conducting a series of iterative studies on the digital departments of 31 museums. Initially, semi-structured interviews were conducted with M_1, complemented by the collection of relevant policy documents, yearbooks, and work summaries. The interview guide was refined based on the findings from M_1, allowing for a more in-depth and efficient survey of M_2, and subsequently from M_3 through M_{31}. Data was meticulously organized following each interview. The final interview guide can be found in Appendix 2. To classify the museums' innovation strategies as either exploitative or exploratory, three researchers independently analyzed the data, followed by group discussions to reach a consensus. In instances where case classification was challenging, follow-up interviews with the museums were conducted.

Targeted data collection. This phase involved collecting measurement data essential for fsQCA, including innovation agreements and financial statements. Additionally, it required collecting representative case data that reflect the innovation configurations identified through QCA analysis, in order to elucidate the causal mechanisms underlying innovation strategies.

Additionally, we employed an iterative validation process with museum managers, which included multiple rounds of presenting findings, gathering feedback, engaging in discussions, making revisions, and seeking additional perspectives. This comprehensive methodology significantly enhances the reliability and practical applicability of our findings.

Table 1
Data collection summary.

Source of data	Type of data	Content of data	Quantity of data
Primary data	Interview data	Interviews with digitalization directors and staff from 31 museums: 30–240 min per museum, total duration: 22.5 h	280 thousand words
	Internal museum documents	Internal documents provided by museums, e.g., "Smart business decision support management system of shanghai museum"	268 copies
Secondary data	Official museum publications	Yearbooks, news, work summaries, etc., published through official channels such as museum websites, public accounts, Douyin accounts	Data collected from a wide range of sources until no new themes emerged
	Academic materials	a. Relevant academic papers retrieved from CNKI literature database, e.g., "Exploration and practice of museum data visualization: Taking the digitalization of shanghai museum as an example" b. Other research reports, e.g., "Digital transformation of Chinese museums in 2020–2021"	316 copies
	Media materials	News reports published by mainstream Chinese media such as People's Daily, Xinhua News Agency	Data collected from a wide range of sources until no new themes emerged

Table by authors.

4. Study I: identification of key enablers influencing digital innovation in museum tourism

4.1. TOE framework: A comprehensive analytical perspective on enablers

Current research lacks a comprehensive approach to examining the factors influencing digital innovation. To address this gap, this study evaluates several technological innovation frameworks and selects the TOE Framework (Tornatzky & Fleischer, 1990) as the analytical tool for the following reasons: Firstly, the TOE framework has been extensively validated across numerous studies, showcasing its strong explanatory power in analyzing organizational innovation behaviors (Awa, Ojiabo, & Orokor, 2017; Baker, 2012). It has also been effective in explaining digital innovation phenomena in tourism destinations (Wang, Li, Li, & Zhang, 2016). Secondly, previous research in museum tourism has primarily focused on technological factors, with organizational factors receiving less emphasis and environmental factors being largely overlooked (Poutanen et al., 2016). In contrast, the comprehensive nature of the TOE framework offers a balanced perspective that integrates technological, organizational, and environmental dimensions, making it particularly well-suited for analyzing complex digital innovation phenomena (Baker, 2012). Additionally, the TOE framework's high degree of flexibility allows it to be adapted to various contexts, thereby enhancing its applicability in diverse innovation analysis scenarios (Awa et al., 2017).

4.2. Data analysis

Study I integrates deductive (top-down) guidance from the TOE framework with inductive (bottom-up) qualitative data analysis. This analysis is conducted through the coding of case materials and follows these steps: Firstly, initial data are assigned conceptual labels to abstract first-order concepts. Secondly, first-order concepts related to technology, organization, and environment are filtered and integrated to develop second-order enablers. Lastly, through comparative analysis, core second-order enablers relevant to both exploitative and exploratory innovation, which are universally significant across all cases, are identified (see Appendix 3).

In addition, it is crucial to manage the number of core second-order enablers to provide a parsimonious explanation when they are included as antecedent conditions in configurational analysis (Furnari et al., 2021). Ragin (2009) suggests that the optimal number of configurational conditions for a medium-sized sample (10–40 cases) ranges from 4 to 7. Ultimately, 6 s-order enablers were identified: cultural relics data assets, smart tourism systems, dynamic capabilities, innovation alliances, public demand, and financial backing.

4.2.1. Technological dimension

Current research on digital innovation in museum tourism places significant emphasis on technologies for visitor interaction (refer to Section 2.2). However, it is crucial to examine the fundamental technological underpinnings of digital innovation. According to Yoo et al. (2010), the characteristics of digital technology—namely data homogeneity, reprogrammability, and convergence—suggest that digital innovation is inherently a data-knowledge-driven process, which

differentiates it significantly from traditional innovation paradigms (Lu, Moyle, et al., 2023; Xu et al., 2024). Analysis results indicate that cultural relics data assets and smart tourism systems, representing comprehensive manifestations of artifact data and tourism-related data along with their associated technologies, constitute the technological bedrock for the sustainable development of museum digital innovation.

Cultural relics data assets, comprising digitized collections and archives, can be integrated with various technologies to facilitate digital content creation and digital dissemination (Liu, 2022; Pesce et al., 2019). A museum representative noted, *"Each digital product and service emerges from the deconstruction of digital cultural resources, subsequently applying digital technology to develop a new narrative framework for exhibitions."* Consequently, both exploitative and exploratory innovation activities require cultural relics data assets as foundational resources.

Second, smart tourism systems are management and decision-support systems developed through the integrated analysis of multimodal data concerning visitors, venues, and operations (Hamid et al., 2021; Wang et al., 2023). These systems intelligently identify visitor patterns to support exploitative innovations in visitor experiences (Giannini & Bowen, 2022) and predict innovation opportunities to foster personalized experiences in exploratory innovations (Cuomo et al., 2021; Johnstone et al., 2023; Torres-Ruiz et al., 2020). For instance, the Jiangxi Provincial Museum has transitioned from passive to proactive innovation through the implementation of the Smart Jiangxi Museum system.

4.2.2. Organizational dimension

The organizational dimension emphasizes essential capabilities that are integral to digital innovation activities. From a complex innovation systems perspective, the aggregation of knowledge and technology from multiple stakeholders is crucial for addressing innovation challenges (Dougherty, 2017; Garud et al., 2011). Consequently, both internal and external organizational capabilities are essential for museums. The analysis underscores the importance of dynamic capabilities and innovation alliances, highlighting the museum's ability to adapt dynamically and the collaborative innovation capacity of external partners, respectively (Warner & Wäger, 2019; Xia et al., 2024).

Dynamic capabilities refer to a museum's ability to sense, integrate, and reconfigure resources to adapt flexibly to changing tourism demands (Teece, 2007). These capabilities are closely linked to both exploitative and exploratory innovation (Limaj & Bernroider, 2019). In today's more diverse and volatile visitor market, understanding visitor needs accurately is crucial for capitalizing on innovation opportunities (Warner & Wäger, 2019). Furthermore, the effective integration and reconfiguration of resources—such as artifacts, technology, talent, and collaborative efforts—are essential for fostering innovative activities (Warner & Wäger, 2019).

Second, as non-commercial institutions, museums often face challenges in independently identifying and addressing the diverse and evolving needs of visitors (García-Muiña et al., 2019; Tim et al., 2020). Innovation alliances, however, offer substantial digital benefits, including deeper market insights, the integration of cutting-edge technologies, personalized audience engagement, and advanced new media marketing techniques (Li & Ghirardi, 2019; Taormina & Baraldi, 2022; Xia et al., 2024). These alliances support both exploitative and exploratory innovations, thereby alleviating the innovation pressures on museum organizations (Tim et al., 2020).

4.2.3. Environmental dimension

The environmental dimension encompasses the policy, economic, market, and cultural factors influencing digital innovation (Baker, 2012; Poutanen et al., 2016). The analysis highlights the crucial role of public demand and financial backing, which museum managers recognize as key determinants in shaping digital innovation strategies.

Firstly, the public-centered nature of museum tourism indicates that public demand is a primary driver of digital innovation (Giannini & Bowen, 2022). The nature and scale of public demand significantly influence innovation strategies in various ways (Johnstone et al., 2023). For instance, public demand data can indicate visitor satisfaction and revisit intention, which influence managerial decisions on whether to continue with exploitative innovation of existing experiences or to discontinue them (Zollo et al., 2022). Conversely, when public demand remains consistently low, there may be a pressing need to explore new digital strategies to overcome bottlenecks in tourism attractiveness (Tham et al., 2023).

Secondly, sufficient financial support is essential for both exploitative and exploratory digital innovation activities (Johnstone et al., 2023; Li & Ghirardi, 2019). In China, museum operations are predominantly funded by government subsidies (Xu, Zhang, et al., 2024), with provincial museums receiving additional budget allocations specifically for supporting innovation. All case museums emphasized the importance of public funding, with some explicitly stating, *"Funding issues are the decisive factor affecting the progress of digitalization."*

4.3. Analytical framework for the impact of digital innovation configurations on museum tourism attractiveness

The preceding analysis has elucidated the technological, organizational, and environmental enablers influencing both exploitative and exploratory digital innovation strategies within museums. Nevertheless, innovation strategies in complex systems are shaped by multi-level interactions among these enablers, rather than by isolated factors. Existing research sporadically addresses these interaction relationships. For instance, numerous studies highlight that while technology can directly influence and foster innovation, its effective implementation requires integration with organizational elements to be fully perceived and realized (Camarero et al., 2011; Tim et al., 2020). Moreover, various interactions exist between organizational and environmental factors (Leoni & Cristofaro, 2022). For example, Johnstone et al. (2023) found that the scale of public demand correlates with museum performance, potentially encouraging managers to adopt technology more proactively. Additionally, significant public funding can stimulate innovation within management; however, it may also reduce the incentive to establish external innovation alliances (Camarero et al., 2011).

Despite the insights offered by existing studies, the research remains fragmented and lacks a systematic and clear articulation of how technological, organizational, and environmental enablers interact to influence exploitative or exploratory innovation strategies. Therefore, this study employs the six enablers as antecedent conditions in fsQCA to develop a configurational analysis framework (Fig. 1). This framework aims to elucidate how TOE enablers collaboratively shape distinct exploitative and exploratory digital innovation strategies, thereby enhancing tourism attractiveness.

5. Study II: configurational analysis of digital innovations in museum tourism

5.1. Applicability of the QCA method

Qualitative Comparative Analysis (QCA) (Ragin, 1987) conceptualizes cases as configurations of conditions and employs comparative analysis at the case level to uncover complex causal relationships between these configurations and outcomes. The appropriateness of using QCA for this study is supported by several factors: Firstly, QCA is adept at elucidating the interactions between digital innovation enablers and their influence on innovation strategies (Ragin, 2009). Secondly, QCA adheres to the principle of equifinality, indicating that different configurations of antecedent conditions can result in the same outcome (Ragin, 2009). This principle facilitates the identification of various innovation models, thereby providing museums with tailored practical guidance. Finally, QCA evaluates the consistency of causal configurations with outcomes (Fiss, 2011), thus identifying key innovation

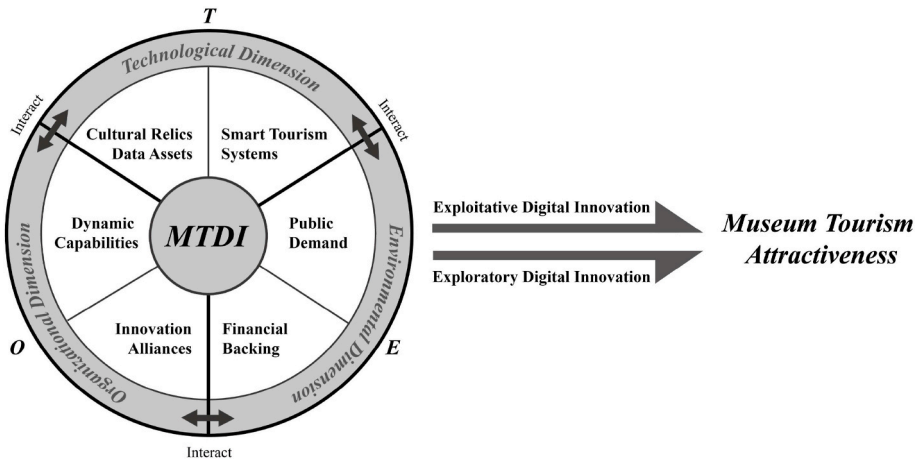

Fig. 1. Analysis model of museum tourism digital innovation (MTDI) influencing museum tourism attractiveness.

strategies with significant practical value. Given that this study utilizes fuzzy set data, the Fuzzy-set Qualitative Comparative Analysis (fsQCA) method is recommended.

5.2. Contrarian case analysis

Contrarian case analysis is an independent preliminary procedure, distinct from QCA. It evaluates the number of cases in the sample that are not explained by the main effects, thereby justifying the use of QCA (Pappas & Woodside, 2021). Following the approach outlined by Pappas and Woodside (2021), we first apply the quintile method to divide the sample variables into five equal groups. Next, we perform a cross-tabulation analysis on these quintiles, as detailed in Appendix 4. The results reveal complex relationships between variables that deviate from main effects, underscoring the necessity of conducting a QCA analysis.

5.3. Measures and calibration

5.3.1. Measurement of antecedent conditions

Cultural relics data assets refer to information about artifacts collected and archived using digital technologies. In this study, it is quantified by the total volume of artifact digitization reported by each museum for the year 2022, as documented in the National Museum Annual Report Information System of the National Cultural Heritage Administration of China. This system adheres to established authoritative norms and statistical standards for the collection, storage, aggregation, and management of artifact information in China.

Smart tourism systems refer to the tourism management and decision-support systems of museums, which emphasize the comprehensive management and utilization of tourism-related data. The Data Management Capability Maturity Model (DCMM) serves as the national standard in China's data management field, categorizing data management maturity into five levels: Initial, Managed, Robust, Quantitatively managed, and Optimized. For the purposes of this study, we adhered to this standard. Since the data management maturity levels of all samples exceed the Initial level, this level was excluded from the analysis. Utilizing the four-value fuzzy set calibration method (Ragin, 2009), we calibrated the data management maturity of museums as follows: a. Managed level (0): Acknowledges the importance of data, incorporating preliminary management processes. b. Robust level (0.33): Views data as a critical performance metric, with standardized management processes. c. Quantitatively managed level (0.67): Considers data a competitive advantage resource, utilizing quantitative analysis. d. Optimized level (1): Treats data as a core resource, featuring continuously optimized management processes and sharing of industry best practices. We systematically evaluated the data management maturity of all museums using primary interview data and secondary sources to assess the implementation level of smart tourism systems.

Dynamic capabilities refer to a museum's ability to sense, integrate, and reconfigure resources in response to evolving tourism demands, particularly emphasizing the capabilities of sensing and integration (Barreto, 2010). We utilize the methodologies proposed by Teece (2007) and Warner and Wäger (2019) to identify and enumerate relevant activities in the 2022 Annual Report of each museum, focusing on sensing capabilities (e.g., perceiving environmental changes, identifying opportunities) and integration capabilities (e.g., integrating, configuring, and utilizing resources). For instance, the Shanghai Museum illustrates: *"S8: By integrating and analyzing data on visitor behavior, cultural product sales, and media dissemination, we perceive changes in the tourism market. S9: We consolidate artifacts, digital technology, and collaborative efforts to launch the Catch the Spring: Digital Special on Jiangnan Culture online exhibition, thus overcoming the temporal and spatial limitations of the museum."*

Innovation alliances refer to collaborative networks of museums dedicated to digital innovation. We evaluated the level of engagement within these innovation alliances by tallying the number of digital-related cooperation agreements each museum had signed by the end of 2022. Our data collection process consists of three steps: 1) Initial compilation: Conduct an initial count by reviewing secondary sources. 2) Verification: Confirm the implementation status of the agreements through primary interviews. 3) Final validation: Thoroughly cross-check all sources to validate the number of legitimate agreements.

Public demand refers to the level of attention and engagement that the public directs towards the museum. In this study, public demand is measured using the 2022 Baidu Index. The Baidu Index, derived from a comprehensive analysis of extensive search queries and user profiles on Baidu—the most widely used search engine in China—stands as one of the most influential platforms for search statistics in the country.

Financial backing refers to public funding provided to support museum innovation. The survey indicates that the museums participating in this study do not allocate fixed financial support for digital innovation activities. Instead, they flexibly allocate resources from the overall budget based on project needs. This suggests that a larger overall financial scale increases the potential for investment in digital innovation activities. Consequently, this study utilizes the annual financial data from the 2022 Annual Financial Report of each museum to assess financial backing.

5.3.2. Measurement of outcome variable

This study considers museum tourism attractiveness as the outcome variable to measure and compare the effectiveness of digital innovation

in museum tourism. We utilize the authoritative Chinese Museum Trending Searches (CMTS) index, managed by Art Exhibitions China (AEC), to quantify this variable. Each quarter, AEC officially compiles and publishes the total CMTS scores for various museums. This score is derived from a comprehensive evaluation across four major dimensions: the Foundation Index (includes the Permanent Exhibition Index, New Exhibition Index, Visitor Index, and Openness Index); the Behavior Index (includes the Platform Index, Cultural Innovation Index, Release Index, and Education Index); the Communication Index (includes the Baidu Index, WeChat Index, Weibo Index, ByteDance All-Platform Index, Short Video Index, Xinbang Index, and Total Network Reading Index); and the Brand Index (includes the Honor Index, Public Sentiment Index, Trending Search Growth Index, Netizen Attention Index, and Media Attention Index).

The CMTS index comprehensively evaluates tourism attractiveness resulting from digital innovation practices, such as new exhibitions created through the integration and utilization of artifact data resources, cultural creative products developed through visitor preference analysis by smart tourism systems, and media content dissemination driven by innovation alliances. Ultimately, we use the average CMTS score for 2022 to measure museum tourism attractiveness.

5.3.3. Calibration

Calibration involves assigning set membership scores to cases, thereby converting them into values ranging from 0.0 to 1.0 (Ragin, 2009). Indirect calibration broadly categorizes cases based on their membership in the target set, while direct calibration assigns values using three qualitative anchors. The smart tourism systems have already been indirectly calibrated as a four-value fuzzy set. For the other six conditions, we employ direct calibration following Pappas and Woodside (2021), using the 95th, 50th, and 5th percentiles of the sample data as anchors for full membership, the crossover point, and full non-membership, respectively (Table 2).

5.4. Identifying necessary conditions

Necessary conditions are those that must be present for a specific outcome to occur (Ragin, 2009). Consistency is the key criterion for identifying necessary conditions; a condition is considered necessary if its consistency score exceeds 0.9. For this analysis, fsQCA 3.0 software was employed. As illustrated in Table 3, the consistency scores for all conditions examined are below 0.9, indicating that there are no necessary conditions for achieving high level of museum tourism attractiveness through digital innovation.

5.5. Identifying sufficient solutions for museum tourism attractiveness

Configurational analysis examines specific combinations of antecedent conditions that are sufficient to produce a particular outcome. For this analysis, we utilized fsQCA 3.0 software, setting the thresholds for both raw consistency and Proportional Reduction in Inconsistency (PRI) consistency at 0.8. Given the focus on a medium-sized sample, the frequency threshold was established at 1 (Ragin, 2009). Additionally, increasing the thresholds for raw and PRI consistency to 0.85 yielded results consistent with the initial analysis, thereby demonstrating the robustness of our findings.

Table 2
Calibration of conditions.

Conditions and outcomes	Full membership	Threshold	Full non-membership
Museum tourism attractiveness	4.55125	3.6725	2.545
Cultural relics data assets	141025.5	21023	2387
Smart tourism systems	–	–	–
Dynamic capabilities	15.5	7	3
Innovation alliances	26	11	2
Public demand	874	256	74
Financial backing	22644.96	10175.1	2179.88

Table 3
Necessary conditions analysis.

	High level museum tourism attractiveness enabled by digital innovation	
Variables	Consistency	Coverage
Cultural relics data assets	0.781723	0.865108
~ Cultural relics data assets	0.536685	0.523980
Smart tourism systems	0.629353	0.760902
~ Smart tourism systems	0.620025	0.563277
Dynamic capabilities	0.686567	0.764543
~ Dynamic capabilities	0.578980	0.562198
Innovation alliances	0.684702	0.780298
~ Innovation alliances	0.565298	0.538188
Public demand	0.833333	0.906631
~ Public demand	0.503109	0.498767
Financial backing	0.700877	0.826848
~ Financial backing	0.564670	0.522741

The results in Table 4 present three solutions for achieving a high level of museum tourism attractiveness. Accurate labeling of configurations enhances theoretical depth and understanding (Furnari et al., 2021). Based on the findings from the fsQCA analysis, relevant theoretical frameworks, and qualitative data, we label museums that focus on exploitative digital innovation, aimed at optimizing established exhibition experiences, as the *Curators*. Conversely, museums that emphasize exploratory digital innovation, which strives to develop novel offerings and experiences, are labeled the *Creators*. Detailed explanations of these categorizations are provided below.

5.5.1. Solution 1: Extroverted Creator

Configuration S1 is characterized by the presence of cultural relics data assets, innovation alliances, and financial backing as core conditions. Museums exemplifying this configuration include the Shanghai Museum, Nanjing Museum, Henan Museum, and Shaanxi

Table 4
Sufficient solutions analysis.

	The digital innovation configurations for achieving a high level of museum tourism attractiveness			
Configurations	S1	S2	S3	
			S3a	S3b
Cultural relics data assets	●	•	⊗	⊗
Smart tourism systems		⊗	●	●
Dynamic capabilities		●	●	●
Innovation alliances	●		⊗	•
Public demand	•	●	•	⊗
Financial backing	●	•	•	•
Consistency	0.9740	0.9743	0.9949	0.9945
Raw coverage	0.4894	0.3532	0.2407	0.2264
Unique coverage	0.1399	0.0417	0.0305	0.018
Solution consistency	0.9784			
Solution coverage	0.5902			

Note: ● indicates the presence of core conditions; ⊗ indicates the absence of core conditions; • indicates the presence of peripheral conditions; ⊗ indicates the absence of peripheral conditions; blank indicates conditions that are optional or irrelevant.

History Museum. These institutions are notable for their historical significance, architectural scale, and the rarity and richness of their collections. Recognized as major museums by the state, they are entrusted with spearheading the creative reinterpretation and innovative evolution of Chinese civilization. They underscore the necessity of having *"the courage to break and reshape. Employing new perspectives and forms of expression to design unique visitor experiences aids the museum industry in overcoming the developmental bottleneck in cultural tourism."* To support such pioneering innovation, initiating high-quality, multidimensional digitization of artifact data is identified as a crucial first step (Pesce et al., 2019). They generally believe that, *"The core of digital innovation lies in how data is used to deconstruct artifacts, thereby producing interpretations that offer new narratives distinct from conventional content."* Additionally, a robust foundation of cultural relics data continuously attracts and enhances the capabilities of external innovators (Pesce et al., 2019). This foundation consistently provides competitive advantages in technology, cross-disciplinary knowledge, innovative thinking, and creative value (García-Muiña et al., 2019; Palumbo, 2022; Tim et al., 2020). In this context, cultural relics data assets and innovation alliances form the core of exploratory digital innovation.

Financial backing, another core enabler, plays a crucial role in motivating employees' enthusiasm for innovation (Taormina & Baraldi, 2022; Tham et al., 2023). A digitalization leader emphasized: *"We encourage our teams to innovate boldly by providing* financial support *for new technologies, skills, and projects. This approach is crucial for fostering creativity."* Contrary to the findings of Camarero et al. (2011), public funding does not result in organizational complacency within museums. Instead, it alleviates tensions between heritage preservation and commercialization in the co-creation processes of innovation alliances (Johnstone et al., 2023).

It is notable that a manager stated, *"Precious artifacts are highly sensitive to environmental changes; even minor alterations can cause irreversible damage. Therefore, changes to the physical gallery facilities must be avoided."* Faced with physical space constraints (Tham et al., 2023), museums have increasingly turned to virtual spaces for innovation. For instance, the Henan Museum partnered with Alipay, a ubiquitous app in China, to launch the Let's Archaeologize digital blind box. This initiative enables users to virtually engage in archaeological excavations, thereby deepening their understanding of artifact histories. Although exploring uncharted territories involves inherent risks, developing and delivering emerging content and visitor value offers a distinct competitive advantage (Pesce et al., 2019). Based on the preceding analysis, we designate the exogenously-oriented exploratory digital innovation Configuration S1 as the Extroverted Creator.

5.5.2. Solution 2: Responsive Curator

Configuration S2 is characterized by dynamic capabilities and public demand as core conditions. In this solution, dynamic capabilities and public demand facilitate a positive interaction between the organizational and environmental dimensions, serving as core drivers of innovation. These museums, on the one hand, demonstrate exceptional dynamic capabilities by perceiving and responding to trends in technology, culture, and consumer behavior through resource integration and the strategic use of digital technologies. Given the absence of support from smart tourism systems (this condition is absent), these museums rely more heavily on their inherent dynamic capabilities to monitor changes in the visitor market. For instance, they employ social media interactions, thematic workshops, and discussion forums to gather visitor feedback, thereby continually optimizing their offerings and experiences.

On the other hand, studies underscore a strong connection between high public demand and museums' star attractions. The unique cultural attributes of these precious artifacts contribute to sustained tourism attractiveness (Wang et al., 2024), thereby influencing strategic management decisions (Giannini & Bowen, 2022). As one digital department manager vividly described, *"Visitors come to our museum specifically for the signature pieces. We strive to give our utmost to perfect their experience."* Consequently, it seems more prudent to thoroughly identify and address established visitor needs rather than risk exploring new offerings.

In this configuration, innovation activities are designed to address and optimize pain points in the visitor experience within existing tourism services. Typically, this involves leveraging digital technologies to enhance or expand the exhibition impact of the museum's most prized masterpieces. This approach offers a deeper and more immersive cultural heritage experience, thereby enhancing visitor engagement (Buhalis et al., 2023). For instance, the Hunan Museum has utilized naked-eye 3D effects to enhance the presentation of the Mawangdui artifacts, complemented by a mid-air suspended LED dome. This arrangement offers visitors an immersive experience, allowing them to journey perceptibly through the Mawangdui space-time. Although cautious, this exploitative innovation of existing exhibits substantially enhances the visitor experience and increases the museum's attractiveness (Trunfio et al., 2022). Based on the aforementioned analysis, we designate the exploitative digital innovation model that focuses on meeting visitor needs and optimizing exhibition experiences as the Responsive Curator.

5.5.3. Solution 3: Introverted Creator

Configuration S3 is characterized by the presence of smart tourism systems and dynamic capabilities as core conditions. This configuration emphasizes the synergy between organizational and technological factors, with dynamic capabilities supported by smart tourism systems, distinguishing it from the non-digital support in Configuration S2. On one hand, smart tourism systems improve the capacity to perceive visitor needs, particularly niche demands (Johnstone et al., 2023; Torres-Ruiz et al., 2020), through comprehensive analysis of data collected from venues, visitor behaviors, and social media feedback (Hamid et al., 2021; Wang et al., 2023). Conversely, these systems possess the capability to efficiently manage museum resources, allowing for swift integration and response to demands (Cuomo et al., 2021). For example, the Guangdong Museum utilizes a smart tourism platform to analyze cultural tourism consumption trends, enabling the innovation of its museum experiences (Wang et al., 2023).

Furthermore, the limited cultural relic data assets (this condition is absent), in Configuration S3 result in reduced traditional exhibition attractiveness. Thus, the breakthrough involves merging creative inspiration with digital technology within the confines of limited relic resources to craft unique and novel experiences. This approach allows museums to strategically cater to personalized visitor needs, particularly attracting individuals who are less interested in the cultural aspects themselves (Fernandez-Lores et al., 2022; Leoni & Cristofaro, 2022). As one manager noted, *"The lack of valuable collections naturally makes it challenging to attract professional visitors. Therefore, we focus on appealing to young visitors and families with children by offering curated and specialized tourism experiences."* Studies indicate that these experiences are frequently embedded in newly established digital spaces within the museum, where cultural elements can be more freely reimagined and enriched with educational and entertaining components to enhance engagement (Komarac, Ozretic-Dosen, & Skare, 2020). For instance, the Guangdong Museum crafted an immersive space where visitors could engage with the creation of export paintings, blending virtual and real-world experiences.

Compared to Configuration S1, Configuration S3 also exhibits characteristics of exploratory innovation strategies. However, due to insufficient cultural relics data resources to engage external innovation

stakeholders, it tends towards endogenous exploratory innovation. Consequently, we designate this configuration as the Introverted Creator.

6. Conclusions and discussions

6.1. Research finding

This study examines representative provincial museums from 31 provincial-level administrative regions in mainland China, utilizing case analysis and fsQCA analysis to investigate the phenomenon of digital innovation in museum tourism. The findings indicate that: (1) The key TOE (Technology, Organization, Environment) enablers influencing digital innovation strategies in museum tourism include cultural relics data assets, smart tourism systems, dynamic capabilities, innovation alliances, public demand, and financial backing. (2) No single digital innovation enabler constitutes a necessary condition for achieving high tourism attractiveness (Table 3). Instead, various combinations of enablers give rise to three digital innovation solutions that can achieve high tourism attractiveness (Table 4): Extroverted Creator (exogenously-oriented exploratory digital innovation), Responsive Curator (responsive exploitative digital innovation), and Introverted Creator (endogenously-oriented exploratory digital innovation).

We have developed a visual model of the enablers (Fig. 2) to provide deeper insight into the complex relationships between TOE enablers and exploitative and exploratory digital innovation strategies: (1) All configurations include an organizational enabler as a core condition, highlighting the critical importance of organizational capabilities (Taormina & Baraldi, 2022). Configurations S1 and S2 also feature an environmental enabler as a core condition, underscoring the previously underappreciated importance of environmental enablers. (2) Exploratory innovation configurations (S1 and S3) both feature a technological enabler—cultural relics data assets or smart tourism systems—as a core condition. In contrast, the significance of technological enablers as a core condition in the exploitative innovation configuration (S2) is not evident. This suggests that exploratory innovation relies on the application of data-driven knowledge (cultural relics data or tourism-related data) to develop innovative offerings, whereas exploitative innovation primarily depends on the application of technical knowledge (e.g., VR, AR) to optimize established offerings (Xu et al., 2024). (3) Table 4 indicates that the coverage of exploratory innovation configurations surpasses that of exploitative innovation configurations, suggesting that large museums are evolving from traditional cultural tourism service providers to pioneers in offering new experiences. However, existing studies primarily focus on exploitative solutions (Tham et al., 2023), highlighting a gap in research concerning large, innovative frontier museums. (4) In exploratory innovation configurations, the S1 exogenously-oriented exploratory digital innovation model (Technology-Organization-Environment) depends on supportive environmental conditions, which enable museums to collaborate with external collaborators more flexibly on cross-boundary innovation projects. In contrast, the S3 endogenously-oriented exploratory model (Technology-Organization) shows lower dependency on environmental conditions. This suggests that large museums can continue to pursue exploratory strategies by cultivating autonomous innovation capabilities and utilizing stable internal resources, even in less favorable environments.

6.2. Theoretical contributions

Firstly, as traditionally non-commercial cultural institutions, museums face the strategic dilemma of choosing between steady, incremental improvements and bold, pioneering transformations (Wang et al., 2024). Deciding whether to continuously enhance established offerings and experiences related to cultural relics or to develop new ones catering to niche visitor needs is a significant yet under-researched question. This study incorporates the theories of exploitative and exploratory innovation (March, 1991; O'Reilly & Tushman, 2004) to elucidate the strengths, limitations, and practical applications of two digital innovation models in museum tourism. It offers a comprehensive explanatory framework for understanding the innovation challenges and opportunities faced by cultural institutions in the tourism sector. This research addresses Giannini and Bowen (2022) 's call for the exploration of new theories and frameworks to better manage structural changes, strategic adjustments, and service innovations in museums. Additionally, it extends the application of exploitative and exploratory innovation classifications in tourism management, as explored by Liu (2018) and Martínez-Pérez, Elche, and García-Villaverde (2021).

Secondly, a crucial method for comprehending the digital innovation strategies of museums involves elucidating the key enablers that facilitate these strategies. However, existing research often lacks a macro perspective in identifying the most critical enablers (Tham et al., 2023). Furthermore, the characteristics of digital technology—specifically data homogeneity, reprogrammability, and convergence (Yoo et al., 2010)—suggest that digital innovation is a data-knowledge-driven process, distinct from traditional innovation models (Xu et al., 2024). Despite this, previous studies have predominantly concentrated on single-instance innovation actions involving the application of technologies, such as AR (García-Muiña et al., 2019), neglecting the potential for digital innovation to continuously evolve. To address these gaps,

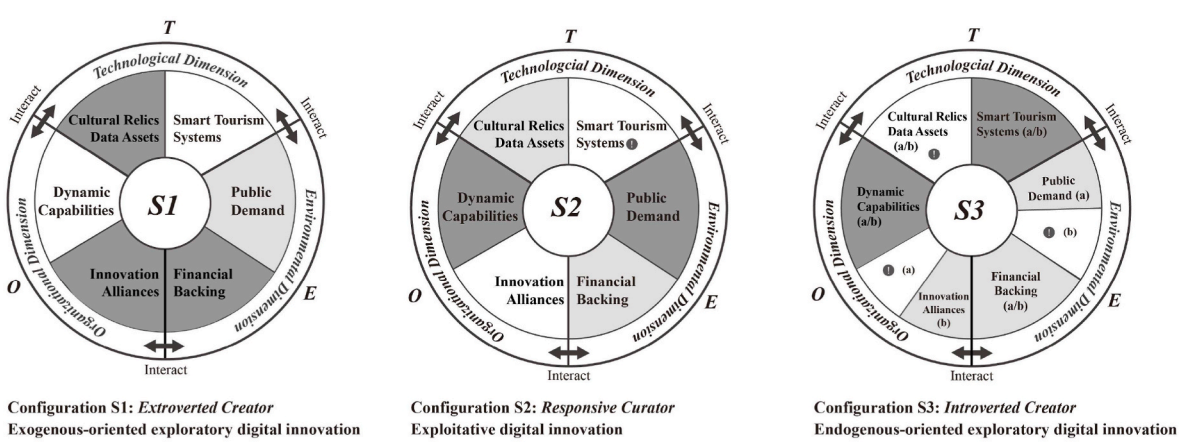

Fig. 2. Visual models for museum tourism digital innovation configurations. Legend: The dark grey areas indicate the presence of core enablers; light grey areas indicate the presence of peripheral enablers; the white areas with exclamation marks indicate a lack of peripheral enablers; blank areas indicate enablers that are optional. Letters in brackets indicates which part of the overlapping solution the enablers belong to.

this study identifies the key enablers of digital innovation, emphasizing cultural relics data assets and smart tourism systems as core technological conditions. This approach advances the application of digital innovation theory within the context of museum tourism. Moreover, previous research has predominantly concentrated on technological and organizational factors, often neglecting the importance of adaptive environmental factors in innovation practices (Poutanen et al., 2016). By incorporating the Technology-Organization-Environment framework, this study offers a comprehensive and relatively balanced assessment of the enablers.

Finally, the perspective of complex innovation systems underscores that innovation emerges from the interplay of multiple enablers (Dougherty & Hardy, 1996; Garud et al., 2011; Poutanen et al., 2016). Previously, Wang et al. (2024) identified three innovation pathways in cultural heritage tourism destinations through multi-level enabler interactions: enterprise-driven, government-guaranteed, and publicly supported. However, there is a notable lack of research exploring the relationship between innovation enablers and exploitative and exploratory strategies. This study employs fsQCA to uncover the pathways of exploitative and exploratory innovation, shaped by the diverse manifestations and interactions of technological, organizational, and environmental enablers in museums. It elucidates the technological and configurational distinctions between these two innovation models, thereby addressing a gap in the existing literature. Furthermore, this study systematically compares the enabler combinations of 31 museums, providing a more comprehensive depiction of the diversity in museum innovation paths compared to previous small-sample case studies (Alshawaaf & Lee, 2021).

6.3. Practical implications

Based on the findings from Study I, museums aiming to pursue digital innovation should prioritize several critical enablers. First and foremost, developing platforms for cultural relics and tourism-related data progressively is crucial to fully leverage the potential of data-driven digital innovation (Xu et al., 2024). Secondly, the internal digital dynamic capabilities of museums, in conjunction with external innovation alliances, collectively form the foundation of their organizational innovation capacity. Enhancing the former involves developing staff's data proficiency and fostering digital creativity. Strengthening the latter requires partnerships with cross-boundary innovation partners. Additionally, evaluating environmental enablers, particularly funding and public demand, is crucial to ensure the feasibility of these innovation strategies.

Based on the findings from Study II, the development of digital innovation strategies is rooted in a thorough evaluation of technological, organizational, and environmental enablers. We developed a digital innovation strategy positioning map for museum tourism (Fig. 3), derived from the configuration analysis of 31 museums, which outlines three primary innovation strategies: (1) Museums with substantial cultural relics data assets can pursue exploratory innovation by deconstructing and reconstructing artifact data. The S1 configuration indicates that this process is significantly enhanced by robust innovation alliances. By integrating the cultural expertise of museum staff (Liu, 2022) with the technological and cross-disciplinary knowledge, along with the innovative thinking of these alliances (García-Muiña et al., 2019), innovative approaches to value creation can emerge. (2) Referencing configuration S2, for museums featuring crown jewel artifacts and experiencing high public demand, exploitative innovation can be leveraged to enhance their attractiveness. The primary focus of innovation is to dynamically identify pain points in the visitor experience and address them through the application of digital technologies. For example, display technologies can extend exhibition spaces and enhance the storytelling of artifacts, offering a more relaxed and enjoyable visitor experience as opposed to a crowded one (Buhalis et al., 2023). (3) For museums without widely recognized collections, exploratory innovation should focus on overcoming limitations in attractiveness. According to the S3 configuration, an effective approach involves leveraging smart tourism systems to monitor cultural consumption trends and gain insights into opportunities for innovation (Hamid et al., 2021). These insights should be combined with the museum's historical narratives to develop new offerings and experiences.

Although innovation pathways are inherently complex and varied (Garud et al., 2011), our commitment lies in broadening the applicability of these solutions across diverse cultural and geographical contexts. Firstly, these strategies have undergone testing and refinement in a range of museums beyond the initial study sample, further validating their applicability. Secondly, the three strategies are inherently adaptable and not mutually exclusive; they can be optimized or restructured to develop new and innovative solutions. Finally, the conclusions of this study are supported by global literature. One exception is that existing research has yet to fully explore the data-driven nature of digital innovation, although this aspect is already being applied in practice by globally renowned museums. For instance, the Rijksmuseum in the Netherlands has launched the Rijksstudio digital art platform to reimagine its collections, while the British Museum and the Louvre have established visitor big data platforms to support innovation in their exhibitions. Moreover, advancements in artificial intelligence and big data analytics are poised to significantly enhance the extraction of value from cultural relics data and the analysis of visitor behavior, thereby advancing the data-knowledge-driven digital innovation process (Xu et al., 2024). These advancements indicate that the strategies proposed in this study hold significant global relevance and demonstrate foresight in anticipating future trends in digital innovation.

6.4. Research limitations and suggestions for future study

Firstly, further research on digital innovation strategies in museums of varying scales, cultural contexts, and policy environments is needed to enrich the theoretical framework. Secondly, while extensive

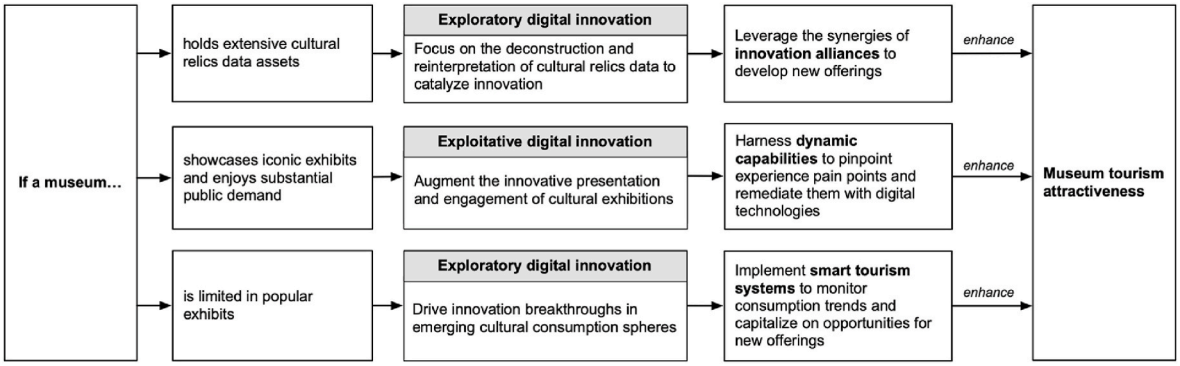

Fig. 3. Museum tourism digital innovation strategy positioning map.

qualitative data was collected to elucidate the characteristics of different innovation configurations and models, more detailed insights should be obtained through in-depth case studies. Lastly, exploring the dynamic balance and transition between exploitative and exploratory innovation (Poutanen et al., 2016) will further develop the ambidextrous innovation theory in the context of museum tourism.

Impact statement

This study examines the enablers and multifaceted mechanisms of digital innovation in museum tourism, making substantial contributions to various aspects of the innovative development of museums. Firstly, the research emphasizes the influence of three dimensions—technology, organization, and environment—on this process. Specifically, it proposes that cultural relics data, tourism-related data, and associated technologies serve as fundamental pillars, enabling museums to harness data-driven knowledge for sustained digital innovation. Secondly, the study provides insights into both exploitative and exploratory digital innovations, guiding museums in strategically choosing between conservative and pioneering approaches. Lastly, the research identifies three digital innovation solutions that enhance the attractiveness of museum tourism and develops a corresponding strategic positioning map. Museum managers can utilize this map to select appropriate innovation strategies or develop new ones through optimization and reorganization.

CRediT authorship contribution statement

Jin Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peihua Shi:** Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Xi Chen:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tourman.2024.105024.

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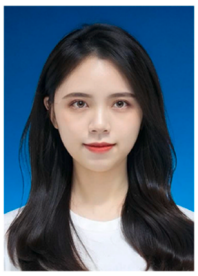

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